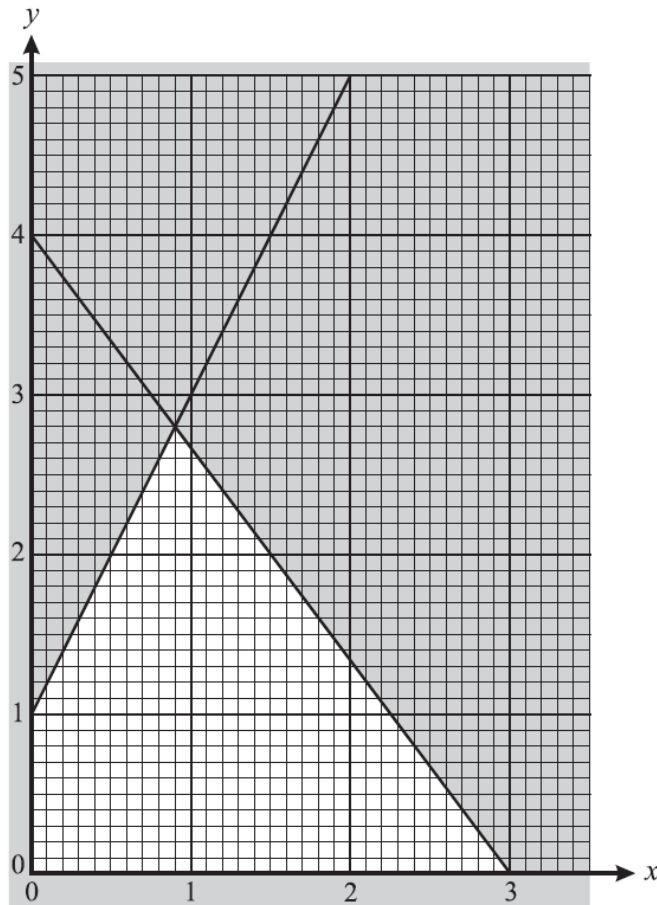


Discrete Revision: Linear programming

- 1 The constraints of a linear programming problem are represented by the graph below. The feasible region is the unshaded region, including its boundaries.



- (a) Write down four inequalities that define the feasible region.

The objective is to maximise $P = 5x + 3y$.

- (b) Using the graph or otherwise, find the maximum value of P and the corresponding values of x and y .

The objective is changed to maximise $Q = ax + 3y$.

- (c) For what set of values of a is the maximum value of Q equal to 3.

- 2 Dane is making the bridesmaid dresses and pageboy outfits for his sister's wedding. He expects it to take 20 hours to make each bridesmaid dress, and 15 hours to make each pageboy outfit. Each bridesmaid dress uses 8 metres of fabric, and each pageboy outfit uses 3 metres of fabric. The fabric costs £8 per metre. Additional items cost £36 for each bridesmaid dress and £96 for each pageboy outfit.

Dane's sister wants to have at least one bridesmaid and at least one pageboy. Dane has 100 hours of time available and must not spend more than £600 in total on materials.

Let x denote the number of bridesmaids and y denote the number of pageboys.

- (a) Show why the constraint $4x + 3y \leq 20$ is needed, and write down three more constraints on the values of x and y , other than that they must be integers.
- (b) Plot the feasible region where all four constraints are satisfied.

Dane's sister wants to maximum the total number of attendants (bridesmaids and pageboys) at her wedding.

- (c) Use your graph to find the maximum number of attendants.
- (d) What is the maximum amount of free time that Dane could have? What is the maximum amount of money that could be left over?

- 3 A company manufactures two types of potting compost, Flowerbase and Growmuch. The weekly amounts produced of each are constrained by the supplies of fibre and of nutrient mix. Each litre of Flowerbase requires 0.75 litres of fibre and 1 kg of nutrient mix. Each litre of Growmuch requires 0.5 litres of fibre and 2 kg of nutrient mix.

Let x and y be the weekly volumes, in thousands of litres, of Flowerbase and Growmuch produced respectively. The feasible values of x and y are shown in the graph below.



- (a) Find the weekly amounts of fibre and nutrient mix available to the company.

The profit on Flowerbase is 9p per litre. The profit on Growmuch is 20p per litre.

- (b) Find the maximum weekly profit the company can make. Give the weekly volumes of Flowerbase and Growmuch that the company should produce.
- (c) Consider each of the following separate circumstances:
- (i) There is a reduction in the weekly supply of fibre to 10,000 litres. What effect does this have on profit?
 - (ii) The price of fibre is increased. Will this affect the optimal production plan? Justify your answer.
 - (iii) The supply of nutrient mix is increased to 30,000 kg per week. What is the new profit?

- 4 A bakery makes three types of doughnut: ring, jam, and custard. The bakery has the following constraints on the number of doughnuts it must make each day:

- The total number of doughnuts made must be at least 200.
- They must make at least twice as many ring doughnuts as jam doughnuts.
- At most 70% of the doughnuts the bakery makes must be ring doughnuts.
- At least a fifth of the doughnuts the bakery makes must be jam doughnuts.

It costs 8 pence to make each ring doughnut, 10 pence to make each jam doughnut, and 14 pence to make each custard doughnut. The bakery wants to minimise the total daily costs of making the required doughnuts.

Let x represent the number of ring doughnuts, let y represent the number of jam doughnuts, and let z represent the number of custard doughnuts the bakery makes each day.

- (a) Formulate this as a linear programming problem, stating the objective and listing the constraints as simplified inequalities with integer coefficients.

One day, instead of making at least 200 doughnuts, the bakery requires that **exactly 200** doughnuts are made.

- (b) Determine the number each type of doughnut the bakery should make on that day.

- 5 The table shows the butter and sugar content in two recipes. The first recipe is for 1 kg of toffee and the second is for 1 kg of fudge.

	Toffee	Fudge
Butter	100 g	150 g
Sugar	800 g	700 g

A confectioner has 1.5 kg of butter, and 10 kg of sugar available. There are no constraints on the availability of other ingredients.

- (a) What is the maximum amount of toffee which the confectioner could make? How much butter or sugar would be left over?

What is the maximum amount of fudge which the confectioner could make? How much butter or sugar would be left over?

- (b) Formulate a linear programming problem to find the maximum total amount of toffee and fudge which the confectioner can make.

Solve your linear programming problem graphically.

The confectioner charges £5.50 for 1 kg of toffee and £4.50 for 1 kg of fudge.

- (c) What quantities should she make to maximise her income? Justify your answer.
- (d) By how much would the price of toffee have to change for the maximum income solution to change?

- 6 The graph below shows the feasible region in a linear programming problem.



The constraints are $x + y \leq 10$, $4x + y \leq 28$, and $2x + 5y \leq 41$.

The objective is to maximise the value of $P = kx + 3y$, where $k \geq 0$.

Find the range of values of k for which (3,7) is an optimal point.